

Release notes: v1.1.0

Repository source: [RELEASE_NOTES.md](#)

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Release date: September 26, 2026 **Author:** Bruce J. Swihart **Status:** AI-assisted, non-peer-reviewed computer-assisted research preprint seeking independent mathematical verification

Canonical repository: github.com/swihart/minvol-degree3-contact-bound

Versioned release: [v1.1.0](#)

Main manuscript: [PDF](#) | [LaTeX source](#)

Purpose of this release

Version **v1.1.0** is a minor theorem-strengthening release of the public computer-assisted universal lower bound in the three-dimensional Blaschke-Lebesgue problem. It preserves the **v1.0.0** proof architecture and replaces only the terminal lower-circumradius row by two independently tuned exact subrows.

For every convex body $K \subset \mathbb{R}^3$ of constant width $d > 0$, the package represents the proposed inequality

$$\text{Vol}(K) \geq \frac{26221074253\pi}{200000000000} d^3 = 0.4118796712122863775432315385981549189 \dots d^3.$$

The result is universal: no symmetry, smoothness, contact-type, or fixed circumradius assumption appears in the final theorem.

Included materials

This release contains:

- a ten-page LaTeX manuscript and compiled PDF;
- a standalone exact-rational certificate;
- immutable JSON, NPZ, TSV, and CSV proof objects;
- an independent NumPy binary64 audit;
- an independent base-R audit;
- ten reconstruction and regression tests;
- seven adversarial proof-object mutation tests;
- automatic paper/certificate and appendix/certificate reconciliation;
- reproducibility, trust-boundary, claims, source, audit, and proof ledgers;
- 22 paired Markdown/PDF documents;

- SHA-256 manifests for the repository, certificate, paper, and rendered docs;
- deterministic release-asset build and verification scripts; and
- a three-job GitHub Actions workflow for Python, R, and PDF/document checks.

Proof architecture

The proof splits the normalized circumradius interval at

$$R_* = \frac{305695601}{500000000} = 0.611391202.$$

The lower branch is a gap-free exact radial certificate. The former terminal row 03B02 is replaced by the exact rows 03B02A and 03B02B; both strictly clear the displayed theorem coefficient. In the near-Jung branch, four circumsphere contacts force a 98,306-ray core. An exact determinant transfer, a dominant-edge coordinate-width estimate, and three optimized antipodal cap pairs give

```
near-Jung lower = 0.4118796712148324317327979810531508741628745212952960018
v1.1 theorem    = 0.4118796712122863775432315385981549188973081955671703077
near-Jung margin over theorem = 0.000000000002546054189566442454995955265...
```

The unchanged near-Jung branch is now the universal bottleneck.

Verification status

Before the final version transition, audited release-candidate commit 892d4bc6fad07950e78da66e45b95063a6415af6 passed:

- the exact finite verifier and byte-identical certificate reconstruction;
- the independent Python binary64 audit;
- the independent base-R audit on the named author's machine and hosted R 4.6.1;
- ten regression tests and seven mutation tests;
- paper/certificate and exact-appendix reconciliation;
- the manuscript and all 22 Markdown/PDF builds;
- hosted Poppler preflight of committed and rebuilt PDFs;
- three green hosted jobs at <https://github.com/swihart/minvol-degree3-contact-bound/actions/runs/35468019663>;
- a complete named-author fresh-clone replay under Python 3.14.7, NumPy 2.3.5, and R 4.6.1; and
- a clean fresh-clone working tree and complete five-commit bundle.

The exact v1.1.0 tag must pass the same three hosted jobs before the GitHub Release is published. Release assets are then built from that tagged checkout, verified against one SHA-256 manifest, downloaded, and reverified in a clean directory.

These checks establish reproducibility of the supplied computations and documents. They do not constitute peer review or external subject-matter verification of the complete mathematical argument.

Release assets

The tagged asset builder produces:

```
minvol-contact-bound-paper-v1.1.0.pdf
minvol-contact-bound-paper-source-v1.1.0.zip
minvol-contact-bound-certificate-v1.1.0.zip
minvol-contact-bound-docs-v1.1.0.zip
minvol-contact-bound-replay-transcripts-v1.1.0.zip
minvol-contact-bound-release-record-v1.1.0.json
RELEASE_NOTES.md
RELEASE_NOTES.pdf
SHA256SUMS.txt
```

Publication metadata

- sole named author: Bruce J. Swihart;
- no institutional affiliation asserted;
- canonical repository: <https://github.com/swihart/minvol-degree3-contact-bound>;
- preferred citation: `CITATION.cff`;
- paper and prose license: CC BY 4.0;
- software and build-infrastructure license: MIT;
- version: `v1.1.0`;
- release date: September 26, 2026;
- archival DOI: none assigned or implied.

Scope and limitations

This release does not prove:

- sharpness or equality;
- rigidity;
- the identity of a minimizing body;
- the Meissner conjecture;
- independent external reproduction or peer review.

The Meissner value is an explicit-body comparison value, not a universal minimum established here. The dated literature check in [proof/SOURCE_LEDGER.md](#) is targeted rather than exhaustive, so this release avoids an unqualified “best known” claim.

Relationship to prior work

The package builds on classical constant-width geometry, Nishioka’s support-function and spectral perspective, and the direct contact-geometric exact-rational architecture of the public HYRA predecessor. The repository has fresh Git history and does not alter the [earlier public MinVol spectral-bound repository](#).

Preferred citation

Bruce J. Swihart, *A Certified Contact-Geometric Lower Bound for the Three-Dimensional Blaschke-Lebesgue Problem*, version `v1.1.0`, computer-assisted preprint, September 26, 2026.

The machine-readable citation is in [CITATION.cff](#).

Corrections and versioning

A change to any theorem-bearing proof object, exact verification rule, radius partition, or coefficient requires a new version and explicit mathematical comparison. The public `v1.0.0` tag and release remain immutable. Corrections must preserve all released history and must not silently rewrite either release.